

Q1 - 24 January - Shift 1

The number of 9 digit numbers, that can be formed using all the digits of the number 123412341 so that the even digits occupy only even places, is ____

Space for your notes:

Q2 - 24 January - Shift 1

A boy needs to select five courses from 12 available courses, out of which 5 courses are language courses. If he can choose at most two language courses, then the number of ways he can choose five courses is

Space for your notes:

Q3 - 24 January - Shift 2

The number of integers, greater than 7000 that can be formed, using the digits 3, 5, 6, 7, 8 without repetition, is

Space for your notes:

- (1) 120
- (2) 168
- (3) 220
- (4) 48

Q4 - 25 January - Shift 1

Let $S = \{1, 2, 3, 5, 7, 10, 11\}$. The number of non-empty subsets of S that have the sum of all elements a multiple of 3, is ____.

Space for your notes:

Q5 - 25 January - Shift 2

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The number of numbers, strictly between 5000 and 10000 can be formed using the digits 1,3,5,7,9 without repetition, is

- (1) 6 (2) 12
(3) 120 (4) 72

Space for your notes:

Q6 - 25 January - Shift 2

Suppose Anil's mother wants to give 5 whole fruits to Anil from a basket of 7 red apples, 5 white apples and 8 oranges. If in the selected 5 fruits, at least 2 orange, at least one red apple and at least one white apple must be given, then the number of ways, Anil's mother can offer 5 fruits to Anil is

Space for your notes:

Q7 - 29 January - Shift 1

If all the six digit numbers $x_1 x_2 x_3 x_4 x_5 x_6$ with $0 < x_1 < x_2 < x_3 < x_4 < x_5 < x_6$ are arranged in the increasing order, then the sum of the digits in the 72th number is _____.

Space for your notes:

Q8 - 29 January - Shift 1

Five digit numbers are formed using the digits 1, 2, 3, 5, 7 with repetitions and are written in descending order with serial numbers. For example, the number 77777 has serial number 1. Then the serial number of 35337 is _____.

Space for your notes:

Q9 - 29 January - Shift 2

The number of 3 digit numbers, that are divisible by either 3 or 4 but not divisible by 48, is

- (1) 472 (2) 432
(3) 507 (4) 400

Space for your notes:

Q10 - 29 January - Shift 2

The letters of the word OUGHT are written in all possible ways and these words are arranged as in a dictionary, in a series. Then the serial number of the word TOUGH is :

- (1) 89 (2) 84
(3) 86 (4) 79

Space for your notes:

Q11 - 29 January - Shift 2

The total number of 4-digit numbers whose greatest common divisor with 54 is 2, is _____.

Space for your notes:

Q12 - 30 January - Shift 1

Number of 4-digit numbers (the repetition of digits is allowed) which are made using the digits 1, 2, 3 and 5, and are divisible by 15, is equal to _____.

Space for your notes:

Q13 - 30 January - Shift 2

The number of ways of selecting two numbers a and b ,
 $a \in \{2, 4, 6, \dots, 100\}$ and $b \in \{1, 3, 5, \dots, 99\}$
 such that 2 is the remainder when $a + b$ is divided
 by 23 is

- (1) 186 (2) 54
 (3) 108 (4) 268

Space for your notes:

Q14 - 30 January - Shift 2

The number of seven digits odd numbers, that can
 be formed using all the seven digits 1, 2, 2, 2, 3, 3, 5
 is _____.

Space for your notes:

Q15 - 31 January - Shift 1

Number of 4-digit numbers that are less than or
 equal to 2800 and either divisible by 3 or by 11, is
 equal to _____.

Space for your notes:

Q16 - 31 January - Shift 1

Let 5 digit numbers be constructed using the digits
 0, 2, 3, 4, 7, 9 with repetition allowed, and are
 arranged in ascending order with serial numbers.
 Then the serial number of the number 42923 is

Space for your notes:

Q17 - 31 January - Shift 2

$$(6A + 11)^2 = 125$$

If ${}^{2n+1}P_{n-1} : {}^{2n-1}P_n = 11 : 21$, then $n^2 + n + 15$ is
 equal to:

Space for your notes:

Q18 - 01 February - Shift 1

The number of words, with or without meaning, that can be formed using all the letters of the word ASSASSINATION so that the vowels occur together, is _____.

Space for your notes:

Q19 - 01 February - Shift 2

Number of integral solutions to the equation $x + y + z = 21$, where $x \geq 1$, $y \geq 3$, $z \geq 4$, is equal to _____.

Space for your notes:

Q20 - 01 February - Shift 2

The total number of six digit numbers, formed using the digits 4, 5, 9 only and divisible by 6, is _____.

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (60)

Q2 (546)

Q3 (2)

Q4 (43)

Q5 (4)

Q6 (6860)

Q7 (32)

Q8 (1436)

Q9 (2)

Q10 (1)

Q11 (3000)

Q12 (21)

Q13 (3)

Q14 (240)

Q15 (710)

Q16 (2997)

Q17 (45)

Q18 (50400)

Q19 (105)

Q20 (81)

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Q1 (60)

Even digits occupy at even places

$$\frac{4!}{2!2!} \times \frac{5!}{2!3!} = \frac{24 \times 120}{4 \times 12} = 60$$

Q2 (546)

For at most two language courses

$$= {}^5C_2 \times {}^7C_3 + {}^5C_1 \times {}^7C_4 + {}^7C_5 = 546$$

Q3 (2)

Four digit numbers greater than 7000

$$= 2 \times 4 \times 3 \times 2 = 48$$

Five digit number = 5! = 120**Total number greater than 7000**

$$= 120 + 48 = 168$$

Q4 (43)

Elements of the type $3k = 3$

Elements of the type $3k + 1 = 1, 7, 9$

Elements of the type $3k + 2 = 2, 5, 11$

Subsets containing one element $S_1 = 1$

Subsets containing two elements

$$S_2 = {}^3C_1 \times {}^3C_1 = 9$$

Subsets containing three elements

$$S_3 = {}^3C_1 \times {}^3C_1 + 1 + 1 = 11$$

Subsets containing four elements

$$S_4 = {}^3C_3 + {}^3C_3 + {}^3C_2 \times {}^3C_2 = 11$$

Subsets containing five elements

$$S_5 = {}^3C_2 \times {}^3C_2 \times 1 = 9$$

Subsets containing six elements $S_6 = 1$

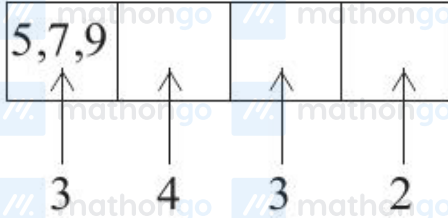
Subsets containing seven elements $S_7 = 1$

$$\Rightarrow \text{sum} = 43$$

Q5 (4)

Numbers between 5000 & 10000

Using digits 1, 3, 5, 7, 9



$$\text{Total Numbers} = 3 \times 4 \times 3 \times 2 = 72$$

Q6 (6860)

7 Red apple(RA), 5 white apple(WA), 8 oranges (O)

5 fruits to be selected (Note:- fruits taken different)

Possible selections :- (2O, 1RA, 2WA) or (2O,

2RA, 1WA) or (3O, 1RA, 1WA)

$$\Rightarrow {}^8C_2 {}^7C_1 {}^5C_2 + {}^8C_2 {}^7C_2 {}^5C_1 + {}^8C_3 {}^7C_1 {}^5C_1$$

$$\Rightarrow 1960 + 2940 + 1960$$

$$\Rightarrow 6860$$

Q7 (32)

1	2				
---	---	--	--	--	--

$$= {}^7C_4 = 35$$

1	3				
---	---	--	--	--	--

$$= {}^6C_4 = 15$$

1	4				
---	---	--	--	--	--

$$= {}^5C_4 = 5$$

1	5				
---	---	--	--	--	--

$$= {}^4C_4 = 1$$

2	3				
---	---	--	--	--	--

$$= {}^6C_4 = 15$$

71 words

2 4 5 6 7 8 $\rightarrow$ 72th word

$$2 + 4 + 5 + 6 + 7 + 8 = 32$$

Q8 (1436)

No of 5 digit numbers starting with digit 1

$$= 5 \times 5 \times 5 \times 5 = 625$$

No of 5 digit numbers starting with digit 2

$$= 5 \times 5 \times 5 \times 5 = 625$$

No of 5 digit numbers starting with 31

$$= 5 \times 5 \times 5 = 125$$

No of 5 digit numbers starting with 32

$$= 5 \times 5 \times 5 = 125$$

No of 5 digit numbers starting with 33

$$= 5 \times 5 \times 5 = 125$$

No of 5 digit numbers starting with 351

$$= 5 \times 5 = 25$$

No of 5 digit numbers starting with 352

$$= 5 \times 5 = 25$$

No of 5 digit numbers starting with 3531

$$= 5$$

No of 5 digit numbers starting with 3532

$$= 5$$

Before 35337 will be 4 numbers,

So rank of 35337 will be 1690

So, in descending order serial number will be

$$3125 - 1690 + 1 = 1436$$

Q9 (2)

Total 3 digit number = 900

Divisible by 3 = 300 (Using $\frac{900}{3} = 300$)

Divisible by 4 = 225 (Using $\frac{900}{4} = 225$)

Divisible by 3 & 4 = 108,

(Using $\frac{900}{12} = 75$)

Number divisible by either 3 or 4

= 300 + 225 - 75 = 450

We have to remove divisible by 48,
144, 192,, 18 terms

Required number of numbers = 450 - 18 = 432

Q10 (1)

Lets arrange the letters of OUGHT in alphabetical order.

G, H, O, T, U

Words starting with

G ---- → 4!

H ---- → 4!

O ---- → 4!

T G ---- → 3!

T H ---- → 3!

T O G -- → 2!

T O H -- → 2!

T O U G H → 1!

Total = 89

Q11 (3000)

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N should be divisible by 2 but not by 3

$N = (\text{Numbers divisible by 2}) - (\text{Numbers divisible by 6})$

$$N = \frac{9000}{2} - \frac{9000}{6} = 4500 - 1500 = 3000$$

Q12 (21)

Sol. For number to be divisible by 15, last digit should be 5 and sum of digits must be divisible by 3.

Possible combinations are

1	2	1	5
---	---	---	---

Numbers = 3

2	2	3	5
---	---	---	---

Numbers = 3

3	3	1	5
---	---	---	---

Numbers = 3

1	1	5	5
---	---	---	---

Numbers = 3

2	3	5	5
---	---	---	---

Numbers = 6

3	5	5	5
---	---	---	---

Numbers = 3

Total Numbers = 21

Q13 (3)

$$a \in \{2, 4, 6, 8, 10, \dots, 100\}$$

$$b \in \{1, 3, 5, 7, 9, \dots, 99\}$$

$$\text{Now, } a + b \in \{25, 71, 117, 163\}$$

(i) $a + b = 25$, no. of ordered pairs (a, b) is 12

(ii) $a + b = 71$, no. of ordered pairs (a, b) is 35

(iii) $a + b = 117$, no. of ordered pairs (a, b) is 42

(iv) $a + b = 163$, no. of ordered pairs (a, b) is 19

$$\therefore \text{total} = 108 \text{ pairs}$$

Q14 (240)

Digits are 1, 2, 2, 2, 3, 3, 5

$$\text{If unit digit 5, then total numbers} = \frac{6!}{3!2!}$$

$$\text{If unit digit 3, then total numbers} = \frac{6!}{3!}$$

$$\text{If unit digit 1, then total numbers} = \frac{6!}{3!2!}$$

$$\therefore \text{total numbers} = 60 + 60 + 120 = 240$$

Q15 (710)

$$1000 - 2799$$

Divisible by 3

$$1002 + (n - 1) 3 = 2799$$

$$n = 600$$

Divisible by 11

$$1 - 2799 \rightarrow \left[\frac{2799}{11} \right] = [254] = 254$$

$$1 - 999 = \left[\frac{999}{11} \right] = 90$$

$$1000 - 2799 = 254 - 90 = 164$$

Divisible by 33

$$1 - 2799 \rightarrow \left[\frac{2799}{33} \right] = 84$$

$$1 - 999 \rightarrow \left[\frac{999}{33} \right] = 30$$

$$1000 - 2799 \rightarrow 54$$

$$\therefore n(3) + n(11) - n(33)$$

$$600 + 164 - 54 = 710$$

Q16 (2997)

$${}^2P_6 + {}^3P_6 + {}^4P_6 + {}^5P_6 = 1296$$

$${}^3P_6 + {}^4P_6 + {}^5P_6 + {}^6P_6 = 1296$$

$${}^{40}P_6 + {}^{41}P_6 + {}^{42}P_6 = 216$$

$${}^{420}P_6 + {}^{421}P_6 = 36$$

$${}^4P_2 + {}^2P_2 + {}^2P_6 = 36$$

$${}^4P_2 + {}^2P_6 = 36$$

$${}^4P_2 + {}^2P_6 = 36$$

$${}^4P_2 + {}^2P_6 = 36$$

$${}^4P_2 + {}^2P_6 = 6$$

$${}^4P_2 + {}^2P_6 = 1$$

$${}^4P_2 + {}^2P_6 = 1$$

$${}^4P_2 + {}^2P_6 = 1$$

$$= 2997$$

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Q17 (45)

$$\frac{(2n+1)!(n-1)!}{(n+2)!(2n-1)!} = \frac{11}{21}$$

$$\Rightarrow \frac{(2n+1)(2n)}{(n+2)(n+1)n} = \frac{11}{21}$$

$$\Rightarrow \frac{2n+1}{(n+1)(n+2)} = \frac{11}{42}$$

$$\Rightarrow n = 5$$

$$\Rightarrow n^2 + n + 15 = 25 + 5 + 15 = 45$$

Q18 (50400)

Vowels : A,A,A,I,I,O

Consonants : S,S,S,S,N,N,T

$\therefore$ Total number of ways in which vowels come together

$$= \frac{8!}{4!2!} \times \frac{6!}{3!2!} = 50400$$

Q19 (105)

$${}^{15}C_2 = \frac{15 \times 14}{2} = 105$$

Q20 (81)

Taking single digit $\rightarrow 444444$ $\frac{6!}{6!} = 1$

Taking two digit $\rightarrow$

(4, 5) 444555 (4, 9) 444999

$$\frac{5!}{3!2!} = 10$$

$$\frac{5!}{3!2!} = 10$$

Taking three digit

$$4, 5, 9, 4, 4, 4 \Rightarrow \frac{5!}{3!} = 20$$

$$4, 5, 9, 5, 5, 5 \Rightarrow \frac{5!}{4!} = 5$$

$$4, 5, 9, 9, 9, 9 \Rightarrow \frac{5!}{4!} = 5$$

$$4, 5, 9, 4, 5, 9 \Rightarrow \frac{5!}{2!2!} = 30$$

Total = 81

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